Machine Learning Predicts the irregular Stress Drop in Laboratory Earthquakes Using Passive Seismic Monitoring of Fault Zone Processes

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Background

Machine-learning approaches have become powerful tools for laboratory earthquake studies since acoustic emissions (AE) generated during stick-slip contain information about fault state. Earlier work used Random Forest, followed by XGBoost and more recent deep learning models, to estimate mechanical behavior from acoustic data. However, many previous datasets were relatively repetitive or quasi-periodic. In contrast, the p5986 experiment studied here lies near the frictional instability boundary, producing strongly aperiodic stick-slip cycles and scattered stress drops. This makes p5986 a stringent test for whether passive AE signals can still predict shear stress evolution in irregular laboratory earthquakes.

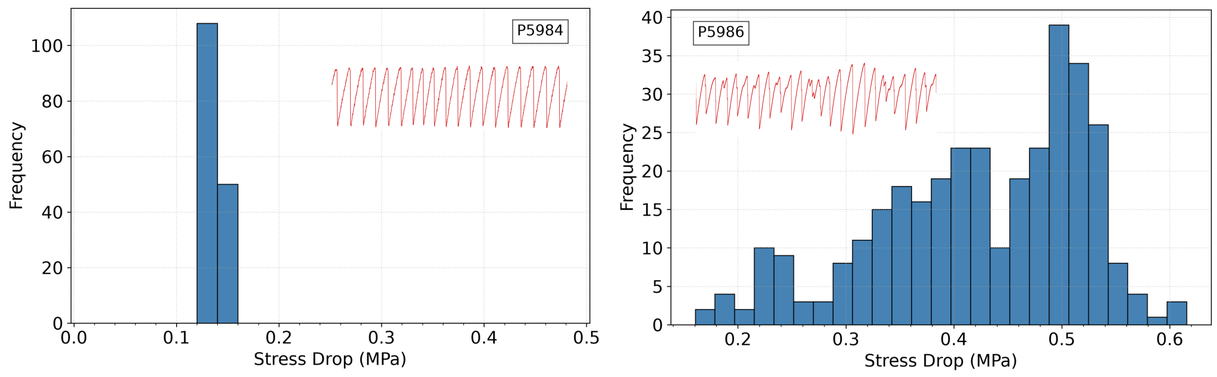


Fig. 1 | Comparison of the stress-drop histograms for p5984 and p5986. p5984 exhibits a periodic distribution in stress drop, whereas p5986 displays a more irregular behavior.

Objectives

- Evaluate whether passive AE signals can predict shear stress evolution in a highly irregular laboratory fault system.
- Compare the predictive capabilities of XGBoost (variance + kurtosis), XGBoost (full features), and LSTM+CNN models.
- Test how model complexity affects performance in a dataset with scattered stress drops and aperiodic stick-slip cycles.

Experiment Configuration

p5986 used a double-direct shear (DDS) configuration with two granular fault-gouge layers between Barre Gray Granite forcing blocks. The analyzed interval was sheared under a constant normal stress of 10 MPa near the frictional instability boundary. Mechanical forces and displacements were sampled at ~1 kHz by DCDT, while passive acoustic emissions were recorded by 12 PZT sensors at ~2.5 MHz using a Verasonics acquisition system.

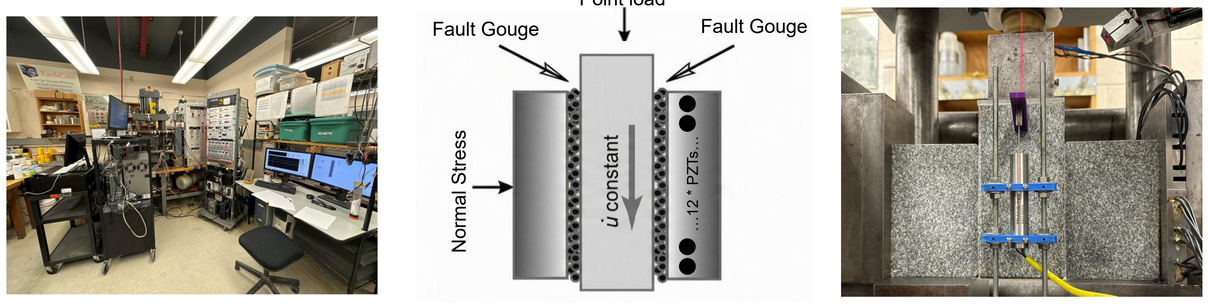


Fig. 2 | Photograph and schematic of p5986.

Methods

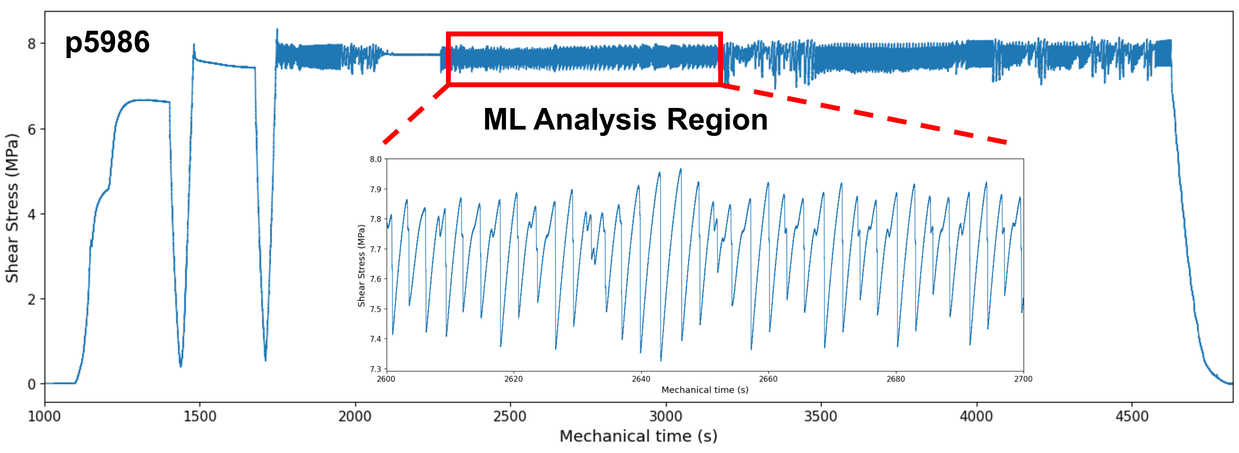


Fig. 3 | Complete shear stress evolution of experiment p5986. An 850-s stick-slip interval (red box) was selected from the p5986 experiment, where stress drops are highly variable. Mechanical and passive acoustic data were synchronized in mechanical time, and a total of 310 failure (stress-drop) events were identified from local shear-stress peaks and subsequent troughs.

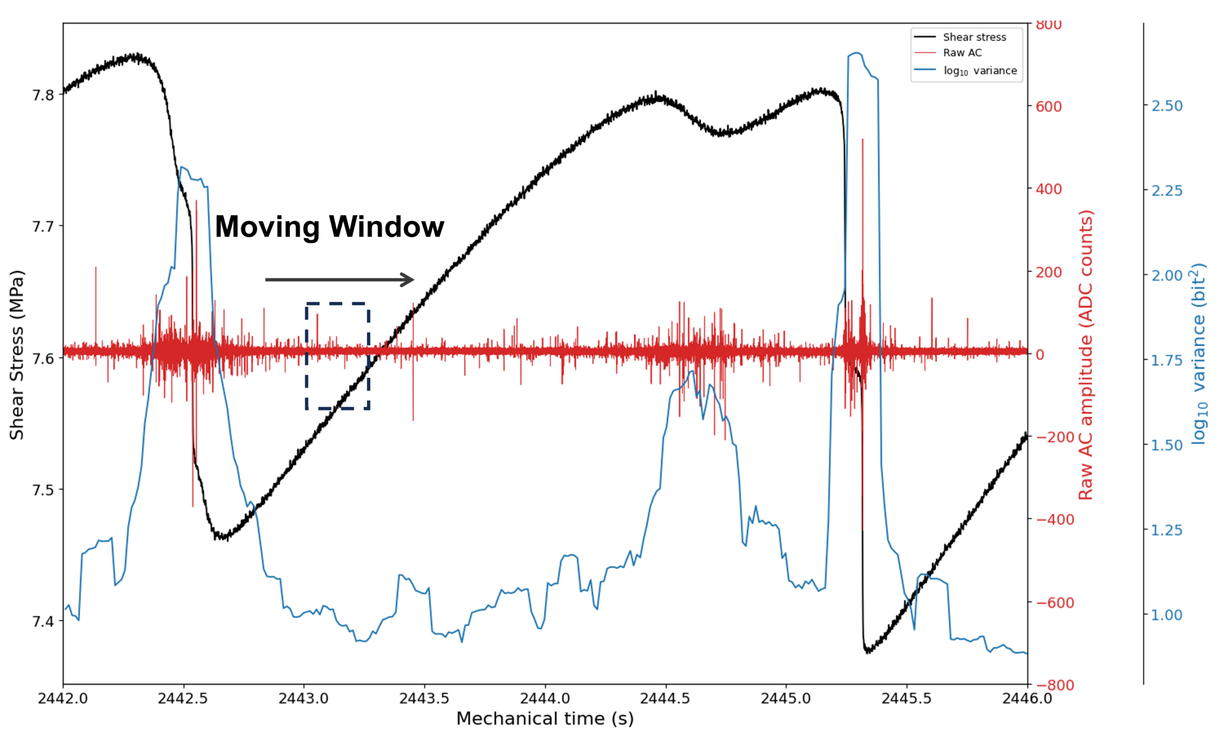


Fig. 4 | Shear stress, acoustic amplitude, and acoustic variance plotted as a function of time for seismic cycles. AC data were processed using moving windows (~0.137 s window, ~0.0137 s step). For each window, acoustic variance was computed as an energy proxy and aligned with the shear-stress record.

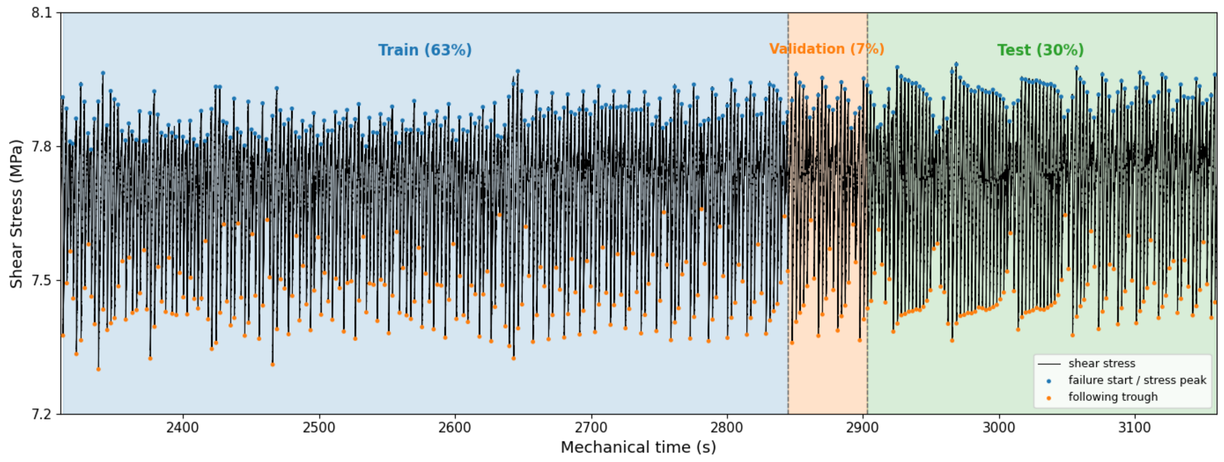


Fig. 5 | Train, validation, and test dataset split with picked stress-drop events. Samples were divided chronologically into train (63%), validation (7%), and test (30%) subsets. We compared XGBoost (variance + kurtosis), XGBoost (full features), and a hybrid LSTM+CNN model for shear-stress prediction.

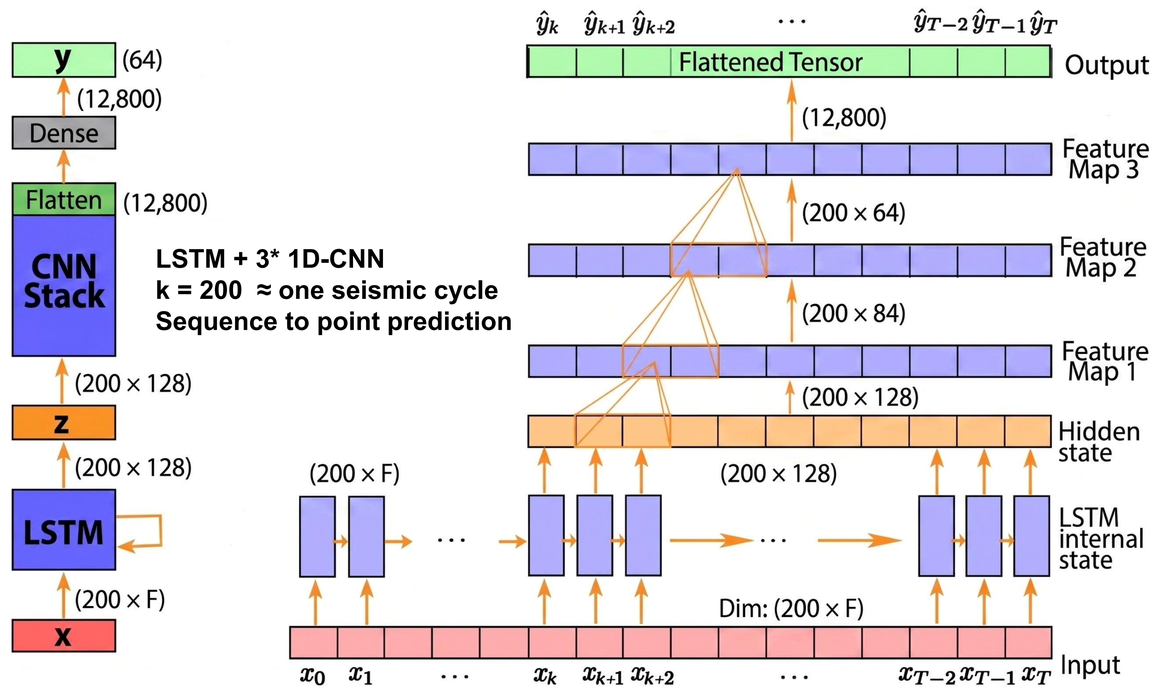
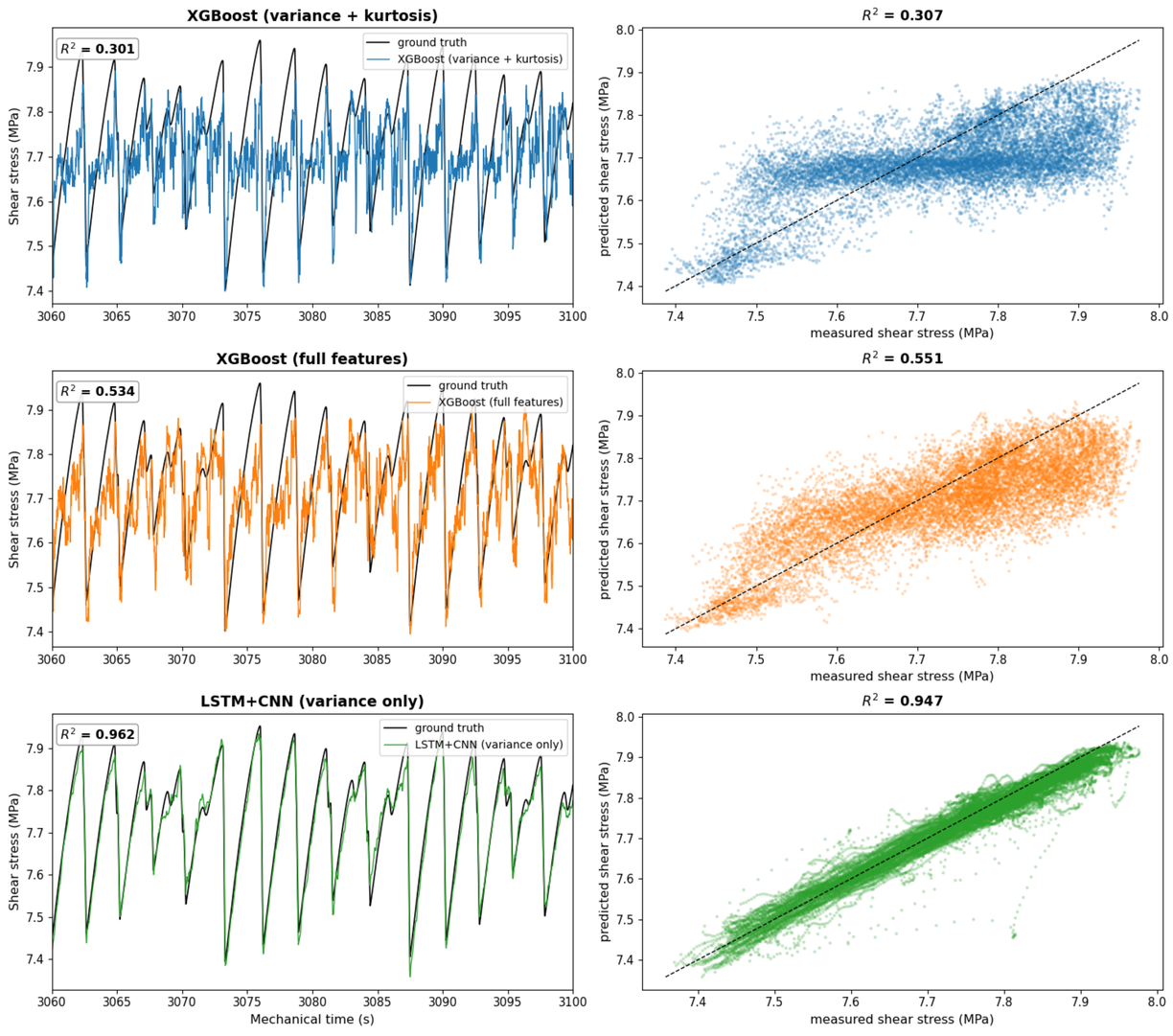


Fig. 6 | Deep Learning Structure : LSTM+CNN for Laboratory Earthquake Prediction.

Results



- Expanding XGBoost from variance + kurtosis to a full statistical and frequency-domain feature set increases test R^2 from 0.307 to 0.551.
- Despite the richer feature set, XGBoost predictions remain compressed and show a banded/step-like prediction structure, indicating limited recovery of continuous shear-stress evolution.
- A LSTM+CNN using only acoustic-variance sequences achieves R^2 =0.947, demonstrating that temporal acoustic structure provides substantially more predictive information than feature expansion alone.

Conclusions

- Passive AE successfully captures the evolving mechanical state of the highly aperiodic p5986 fault system.
- Increasing handcrafted feature complexity improves XGBoost prediction, but temporal sequence modeling provides a substantially larger gain in predictive performance.
- Conventional DDS/SDS experiments remain geometrically simplified relative to spatially heterogeneous natural fault systems.
- Future work will extend passive-acoustic ML toward larger-scale and multi-dimensional laboratory faults, with spatially distributed or sparse sensors.
- Similar approaches will be explored for other geophysical hazards, including debris flows, to test whether system-scale dynamics can be inferred from limited observation points.